# BSAN 360 - 001 - Project

# Youssef Bastawy

In [27]:
```
import statsmodels.api as sm
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from plotly import express as px
import matplotlib.ticker as ticker
```

In [2]:
```
df = pd.read_csv("database_24_25.csv")

significant_minutes_threshold = 10
df_significant_minutes = df[df['MP'] > significant_minutes_threshold]

print("Data loaded successfully.")
```

```
Data loaded successfully.
```

In [3]:
```
print("Minutes Played (MP) statistics:")
print(df['MP'].describe())

if df['MP'].dtype == 'object':

    try:
        df['MP'] = df['MP'].apply(lambda x: float(x.split(':')[0]) + float(x
    except:
        pass

    df['MP'] = pd.to_numeric(df['MP'], errors='coerce')

min_minutes = 30

df_significant = df[df['MP'] >= min_minutes]

print(f"Original dataset: {df.shape[0]} rows, {df.shape[1]} columns")
print(f"After filtering for players with {min_minutes}+ minutes: {df_signifi
print(f"Removed {df.shape[0] - df_significant.shape[0]} rows with less than

numerical_columns = df_significant.select_dtypes(include=['float64', 'int64'
numeric_df = df_significant[numerical_columns]

correlation_matrix = numeric_df.corr()

high_correlation = correlation_matrix[abs(correlation_matrix) > 0.8].stack()
high_correlation.columns = ['col1', 'col2', 'correlation']
high_correlation = high_correlation[high_correlation['col1'] != high_correla

columns_to_remove = high_correlation['col2'].tolist()
```

```
df_final = df_significant.drop(columns=columns_to_remove)

print(f"Final DataFrame shape after removing highly correlated columns: {df_

df = df_final.copy()
print("Updated original df variable with filtered data:")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

df.head()
```

```
Minutes Played (MP) statistics:
count    16512.000000
mean        22.405259
std         10.896645
min          0.000000
25%         14.427500
50%         23.430000
75%         31.322500
max         50.480000
Name: MP, dtype: float64
Original dataset: 16512 rows, 25 columns
After filtering for players with 30+ minutes: 4892 rows, 25 columns
Removed 11620 rows with less than 30 minutes played
Final DataFrame shape after removing highly correlated columns: 4892 rows, 1
7 columns
Updated original df variable with filtered data:
Rows: 4892
Columns: 17
```

Out[3]:

| | Player | Tm | Opp | Res | MP | FG% | 3P | 3PA | 3P% | FT% | ORB | AST | STL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Jayson Tatum | BOS | NYK | W | 30.30 | 0.778 | 8 | 11 | 0.727 | 0.500 | 0 | 10 | 1 |
| **1** | Anthony Davis | LAL | MIN | W | 37.58 | 0.478 | 1 | 3 | 0.333 | 0.867 | 3 | 4 | 1 |
| **3** | Jrue Holiday | BOS | NYK | W | 30.52 | 0.778 | 4 | 6 | 0.667 | 0.000 | 2 | 4 | 1 |
| **5** | Rui Hachimura | LAL | MIN | W | 35.08 | 0.500 | 1 | 4 | 0.250 | 0.750 | 3 | 1 | 2 |
| **7** | Rudy Gobert | MIN | LAL | L | 35.33 | 0.625 | 0 | 0 | 0.000 | 0.750 | 3 | 2 | 0 |

In [4]:

```
print("First 5 rows:")
display(df.head())

print("Last 5 rows:")
display(df.tail())
```

```
First 5 rows:
```

| | Player | Tm | Opp | Res | MP | FG% | 3P | 3PA | 3P% | FT% | ORB | AST | STL | E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Jayson Tatum | BOS | NYK | W | 30.30 | 0.778 | 8 | 11 | 0.727 | 0.500 | 0 | 10 | 1 | |
| **1** | Anthony Davis | LAL | MIN | W | 37.58 | 0.478 | 1 | 3 | 0.333 | 0.867 | 3 | 4 | 1 | |
| **3** | Jrue Holiday | BOS | NYK | W | 30.52 | 0.778 | 4 | 6 | 0.667 | 0.000 | 2 | 4 | 1 | |
| **5** | Rui Hachimura | LAL | MIN | W | 35.08 | 0.500 | 1 | 4 | 0.250 | 0.750 | 3 | 1 | 2 | |
| **7** | Rudy Gobert | MIN | LAL | L | 35.33 | 0.625 | 0 | 0 | 0.000 | 0.750 | 3 | 2 | 0 | |

Last 5 rows:

| | Player | Tm | Opp | Res | MP | FG% | 3P | 3PA | 3P% | FT% | ORB | AST | ST |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **16450** | Carlton Carrington | WAS | CLE | L | 35.82 | 0.556 | 3 | 6 | 0.500 | 0.0 | 1 | 3 | |
| **16465** | Taurean Prince | MIL | ATL | L | 30.82 | 0.571 | 1 | 4 | 0.250 | 0.0 | 0 | 1 | |
| **16467** | Tim Hardaway Jr. | DET | PHI | W | 30.80 | 0.222 | 2 | 8 | 0.250 | 0.0 | 0 | 3 | |
| **16479** | Tyler Herro | MIA | BRK | L | 37.78 | 0.273 | 2 | 11 | 0.182 | 1.0 | 1 | 6 | |
| **16480** | Nikola Jović | MIA | BRK | L | 31.73 | 0.375 | 2 | 5 | 0.400 | 0.0 | 0 | 4 | |

In [5]:
```
print("Original Columns")
print(df.columns.tolist())

df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
    .str.replace("%", "pct")
    .str.replace("/", "_per_", regex=False)
    .str.replace(r"[^a-z0-9_]", "", regex=True)
)

print("Cleaned Columns")
print(df.columns.tolist())

if "team" in df.columns:
    df = df.set_index("team")
    print("Index set to: team")
elif "team_name" in df.columns:
    df = df.set_index("team_name")
    print("Index set to: team_name")
```

```
else:
    print("No team index column found; default index kept.")

display(df.head())
```

```
Original Columns
['Player', 'Tm', 'Opp', 'Res', 'MP', 'FG%', '3P', '3PA', '3P%', 'FT%', 'OR
B', 'AST', 'STL', 'BLK', 'TOV', 'PF', 'Data']
Cleaned Columns
['player', 'tm', 'opp', 'res', 'mp', 'fgpct', '3p', '3pa', '3ppct', 'ftpct',
'orb', 'ast', 'stl', 'blk', 'tov', 'pf', 'data']
No team index column found; default index kept.
```

| | player | tm | opp | res | mp | fgpct | 3p | 3pa | 3ppct | ftpct | orb | ast | stl | blk |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Jayson Tatum | BOS | NYK | W | 30.30 | 0.778 | 8 | 11 | 0.727 | 0.500 | 0 | 10 | 1 | 1 |
| **1** | Anthony Davis | LAL | MIN | W | 37.58 | 0.478 | 1 | 3 | 0.333 | 0.867 | 3 | 4 | 1 | 3 |
| **3** | Jrue Holiday | BOS | NYK | W | 30.52 | 0.778 | 4 | 6 | 0.667 | 0.000 | 2 | 4 | 1 | 0 |
| **5** | Rui Hachimura | LAL | MIN | W | 35.08 | 0.500 | 1 | 4 | 0.250 | 0.750 | 3 | 1 | 2 | 1 |
| **7** | Rudy Gobert | MIN | LAL | L | 35.33 | 0.625 | 0 | 0 | 0.000 | 0.750 | 3 | 2 | 0 | 1 |

In [6]:

```
research_question = (
    "Which statistical categories (shooting efficiency, turnovers, rebounds,
    "best predict winning games?"
)
print("Research Question:")
print(research_question)

if "res" in df.columns:
    df["target_win"] = (df["res"].astype(str).str.upper() == "W").astype(int
elif "Res" in df.columns:
    df["target_win"] = (df["Res"].astype(str).str.upper() == "W").astype(int
else:
    raise ValueError("Could not find result column (res/Res) to build win ta

target = "target_win"
print("Selected target variable:", target)

candidate_predictors = [
    "fg_pct", "fg%", "fg",
    "3p_pct", "3p%", "3p", "3pa",
    "ft_pct", "ft%", "ft", "fta",
    "orb", "drb", "trb",
    "ast", "stl", "blk", "tov", "pf",
    "pts", "gmsc", "mp", "fga"
]

predictors = [c for c in candidate_predictors if c in df.columns]
```

```
print("Predictor columns found:")
print(predictors)

if len(predictors) < 3:
    numeric_cols = df.select_dtypes(include="number").columns.tolist()
    print("Few standard predictors found. Numeric columns available:")
    print(numeric_cols)
```

```
Research Question:
Which statistical categories (shooting efficiency, turnovers, rebounds, et
c.) best predict winning games?
Selected target variable: target_win
Predictor columns found:
['3p', '3pa', 'orb', 'ast', 'stl', 'blk', 'tov', 'pf', 'mp']
```

In [7]:
```
# Target: win_pct (or wins)
# Predictors: shooting (fg_pct, efg_pct, ts_pct), turnovers (tov / tov_pct),
```

In [8]:
```
project_cols = predictors + [target]
nba = df[project_cols].copy()

print("Focused DataFrame shape (before cleaning):", nba.shape)
display(nba.head())

nba = nba.dropna(subset=[target])

for c in predictors:
    if nba[c].isna().any():
        nba[c] = nba[c].fillna(nba[c].median())

print("Focused DataFrame shape (after cleaning):", nba.shape)
print("Any missing values left?")
display(nba.isna().sum())
```

```
Focused DataFrame shape (before cleaning): (4892, 10)
```

| | 3p | 3pa | orb | ast | stl | blk | tov | pf | mp | target_win |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 8 | 11 | 0 | 10 | 1 | 1 | 1 | 1 | 30.30 | 1 |
| **1** | 1 | 3 | 3 | 4 | 1 | 3 | 1 | 1 | 37.58 | 1 |
| **3** | 4 | 6 | 2 | 4 | 1 | 0 | 0 | 2 | 30.52 | 1 |
| **5** | 1 | 4 | 3 | 1 | 2 | 1 | 0 | 2 | 35.08 | 1 |
| **7** | 0 | 0 | 3 | 2 | 0 | 1 | 1 | 4 | 35.33 | 0 |

```
Focused DataFrame shape (after cleaning): (4892, 10)
Any missing values left?
```

```
3p            0
3pa           0
orb           0
ast           0
stl           0
blk           0
tov           0
pf            0
mp            0
target_win    0
dtype: int64
```

In [9]:
```
# Analysis 1: Correlation screen
corr = nba.corr(numeric_only=True)[target].sort_values(ascending=False)
print("Correlation with target:")
display(corr)

top_abs_corr = corr.drop(labels=[target], errors="ignore").abs().sort_values
print("Top predictors by absolute correlation:")
display(top_abs_corr.head(10))

# Analysis 2: Multiple linear regression
X = nba[predictors]
y = nba[target]
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
print(model.summary())
```

```
Correlation with target:
target_win    1.000000
ast           0.084789
blk           0.059775
3p            0.059075
stl           0.053413
mp            0.000770
orb           0.000607
3pa          -0.018616
tov          -0.047921
pf           -0.061306
Name: target_win, dtype: float64
Top predictors by absolute correlation:
ast    0.084789
pf     0.061306
blk    0.059775
3p     0.059075
stl    0.053413
tov    0.047921
3pa    0.018616
mp     0.000770
orb    0.000607
Name: target_win, dtype: float64
```

```
                            OLS Regression Results
==============================================================================
==
Dep. Variable:             target_win   R-squared:                       0.0
40
Model:                            OLS   Adj. R-squared:                  0.0
38
Method:                 Least Squares   F-statistic:                     22.
35
Date:                Thu, 19 Mar 2026   Prob (F-statistic):           1.23e-
37
Time:                        21:44:19   Log-Likelihood:                -345
1.5
No. Observations:                4892   AIC:                             692
3.
Df Residuals:                    4882   BIC:                             698
8.
Df Model:                           9
Covariance Type:            nonrobust
==============================================================================
==
                 coef    std err          t      P>|t|      [0.025      0.97
5]
------------------------------------------------------------------------------
--
const          0.5331      0.075      7.070      0.000       0.385       0.6
81
3p             0.0561      0.006      8.933      0.000       0.044       0.0
68
3pa           -0.0260      0.003     -7.679      0.000      -0.033      -0.0
19
orb           -0.0002      0.005     -0.040      0.968      -0.009       0.0
09
ast            0.0195      0.002      7.933      0.000       0.015       0.0
24
stl            0.0223      0.006      3.649      0.000       0.010       0.0
34
blk            0.0361      0.007      5.089      0.000       0.022       0.0
50
tov           -0.0221      0.004     -5.003      0.000      -0.031      -0.0
13
pf            -0.0203      0.005     -4.013      0.000      -0.030      -0.0
10
mp            -0.0012      0.002     -0.521      0.602      -0.006       0.0
03
==============================================================================
==
Omnibus:                    18833.681   Durbin-Watson:                   2.0
63
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              693.8
71
Skew:                          -0.019   Prob(JB):                     2.13e-1
51
Kurtosis:                       1.155   Cond. No.                         38
8.
==============================================================================
```

```
==

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is corre
ctly specified.
```

In [10]:
```
# I use correlation to screen relationships and multiple linear regression t
```

# Project 2

In [11]:
```
print("DATA CLEANING AND PREPARATION")

print("Initial dataset overview:")
print("Rows: {df.shape[0]}, Columns: {df.shape[1]}")
print("Column names:")
print(df.columns.tolist())

print("Data types:")
print(df.dtypes)

print("Missing values per column:")
print(df.isna().sum())

print("Basic statistics for numerical columns:")
print(df.describe())
```

```
DATA CLEANING AND PREPARATION
Initial dataset overview:
Rows: {df.shape[0]}, Columns: {df.shape[1]}
Column names:
['player', 'tm', 'opp', 'res', 'mp', 'fgpct', '3p', '3pa', '3ppct', 'ftpct',
'orb', 'ast', 'stl', 'blk', 'tov', 'pf', 'data', 'target_win']
Data types:
player         object
tm             object
opp            object
res            object
mp            float64
fgpct         float64
3p              int64
3pa             int64
3ppct         float64
ftpct         float64
orb             int64
ast             int64
stl             int64
blk             int64
tov             int64
pf              int64
data           object
target_win      int64
dtype: object
Missing values per column:
player        0
tm            0
opp           0
res           0
mp            0
fgpct         0
3p            0
3pa           0
3ppct         0
ftpct         0
orb           0
ast           0
stl           0
blk           0
tov           0
pf            0
data          0
target_win    0
dtype: int64
Basic statistics for numerical columns:
                mp        fgpct           3p          3pa        3ppct  \
count  4892.000000  4892.000000  4892.000000  4892.000000  4892.000000
mean     34.864025     0.482701     2.165372     5.766762     0.338116
std       3.286061     0.147828     1.790683     3.464659     0.240041
min      30.000000     0.000000     0.000000     0.000000     0.000000
25%      32.170000     0.385000     1.000000     3.000000     0.167000
50%      34.475000     0.474000     2.000000     6.000000     0.333000
75%      37.030000     0.571000     3.000000     8.000000     0.500000
max      50.480000     1.000000    10.000000    20.000000     1.000000
```

```
             ftpct          orb          ast          stl          blk  \
count  4892.000000  4892.000000  4892.000000  4892.000000  4892.000000
mean      0.643928     1.417212     4.359158     1.171913     0.691128
std       0.383523     1.617065     3.077210     1.159920     1.015550
min       0.000000     0.000000     0.000000     0.000000     0.000000
25%       0.500000     0.000000     2.000000     0.000000     0.000000
50%       0.750000     1.000000     4.000000     1.000000     0.000000
75%       1.000000     2.000000     6.000000     2.000000     1.000000
max       1.000000    12.000000    22.000000     8.000000     9.000000

               tov           pf   target_win
count  4892.000000  4892.000000  4892.000000
mean      2.122445     2.317457     0.505928
std       1.681417     1.405366     0.500016
min       0.000000     0.000000     0.000000
25%       1.000000     1.000000     0.000000
50%       2.000000     2.000000     1.000000
75%       3.000000     3.000000     1.000000
max      11.000000     6.000000     1.000000
```

In [12]:

```
print("HANDLING MISSING DATA")

missing_values = df.isna().sum()
columns_with_missing = missing_values[missing_values > 0].index.tolist()

print("Columns with missing values: {columns_with_missing}")

print("Justification for handling missing data:")
print("For our basketball analysis, we're focusing on key players with signi
print("Players with missing statistics likely had minimal impact on the game
print("Therefore, we'll filter out players with less than 30 minutes of play

df.columns = df.columns.str.strip()

df['mp'] = pd.to_numeric(df['mp'], errors='coerce')

min_minutes = 30

df_filtered = df[df['mp'] >= min_minutes]

print("nOriginal dataset: {df.shape[0]} rows, {df.shape[1]} columns")
print("After filtering for players with {min_minutes}+ minutes: {df_filterec
print("Removed {df.shape[0] - df_filtered.shape[0]} rows with less than {mir

missing_after_filter = df_filtered.isna().sum()
columns_still_missing = missing_after_filter[missing_after_filter > 0].index

print("Columns still with missing values after filtering: {columns_still_mis

if columns_still_missing:
    key_stats = ['PTS', 'FG%', 'TRB', 'AST']
    key_stats_present = [col for col in key_stats if col in df_filtered.colu

    if key_stats_present:
        df_filtered = df_filtered.dropna(subset=key_stats_present)
```

```
        print(f"Dropped rows with missing values in key statistics: {key_sta
        print(f"Remaining dataset: {df_filtered.shape[0]} rows")
```

```
HANDLING MISSING DATA
Columns with missing values: {columns_with_missing}
Justification for handling missing data:
For our basketball analysis, we're focusing on key players with significant
playing time.
Players with missing statistics likely had minimal impact on the game.
Therefore, we'll filter out players with less than 30 minutes of playing tim
e.
nOriginal dataset: {df.shape[0]} rows, {df.shape[1]} columns
After filtering for players with {min_minutes}+ minutes: {df_filtered.shape
[0]} rows, {df_filtered.shape[1]} columns
Removed {df.shape[0] - df_filtered.shape[0]} rows with less than {min_minute
s} minutes played
Columns still with missing values after filtering: {columns_still_missing}
```

In [13]:

```
print("DATA TRANSFORMATIONS")

percentage_cols = [col for col in df_filtered.columns if '%' in col]
print(f"Percentage columns: {percentage_cols}")

for col in percentage_cols:
    if df_filtered[col].dtype == 'object':
        df_filtered[col] = pd.to_numeric(df_filtered[col], errors='coerce')

    if df_filtered[col].max() > 1:
        df_filtered[col] = df_filtered[col] / 100
        print(f"Converted {col} from percentage to decimal format")

print("Creating additional efficiency metrics:")

if all(col in df_filtered.columns for col in ['PTS', 'FGA']):
    df_filtered['PPS'] = df_filtered['PTS'] / df_filtered['FGA'].replace(0,
    print("- Added Points Per Shot (PPS) metric")

if all(col in df_filtered.columns for col in ['AST', 'TOV']):
    df_filtered['AST/TOV'] = df_filtered['AST'] / df_filtered['TOV'].replace
    print("- Added Assist to Turnover ratio (AST/TOV)")

if all(col in df_filtered.columns for col in ['PTS', 'FGA', 'FTA']):
    df_filtered['TS%'] = df_filtered['PTS'] / (2 * (df_filtered['FGA'] + 0.4
    print("- Added True Shooting Percentage (TS%)")

print("Identifying and removing highly correlated features:")

numerical_columns = df_filtered.select_dtypes(include=['float64', 'int64']).
numeric_df = df_filtered[numerical_columns]
correlation_matrix = numeric_df.corr()

high_correlation = correlation_matrix[abs(correlation_matrix) > 0.8].stack()
high_correlation.columns = ['col1', 'col2', 'correlation']
high_correlation = high_correlation[high_correlation['col1'] != high_correla

print("Highly correlated feature pairs (correlation > 0.8):")
```

```
for _, row in high_correlation.iterrows():
    print(f"- {row['col1']} and {row['col2']}: {row['correlation']:.3f}")

columns_to_remove = high_correlation['col2'].tolist()
print(f"Removing {len(columns_to_remove)} highly correlated columns: {columr

df_cleaned = df_filtered.drop(columns=columns_to_remove)
```

```
DATA TRANSFORMATIONS
Percentage columns: []
Creating additional efficiency metrics:
Identifying and removing highly correlated features:
Highly correlated feature pairs (correlation > 0.8):
Removing 0 highly correlated columns: []
```

In [14]:

```
print("FINAL CLEANED DATASET")

print(f"Final dataset dimensions: {df_cleaned.shape[0]} rows, {df_cleaned.sh
print("Remaining columns:")
print(df_cleaned.columns.tolist())

print("Missing values in final dataset:")
print(df_cleaned.isna().sum().sum(), "total missing values")

print("Sample of final cleaned dataset:")
print(df_cleaned.head())

df = df_cleaned.copy()
```

```
FINAL CLEANED DATASET
Final dataset dimensions: 4892 rows, 18 columns
Remaining columns:
['player', 'tm', 'opp', 'res', 'mp', 'fgpct', '3p', '3pa', '3ppct', 'ftpct',
'orb', 'ast', 'stl', 'blk', 'tov', 'pf', 'data', 'target_win']
Missing values in final dataset:
0 total missing values
Sample of final cleaned dataset:
          player   tm  opp res     mp  fgpct  3p  3pa  3ppct  ftpct  orb  as
t  \
0   Jayson Tatum  BOS  NYK   W  30.30  0.778   8   11  0.727  0.500    0   1
0
1  Anthony Davis  LAL  MIN   W  37.58  0.478   1    3  0.333  0.867    3
4
3   Jrue Holiday  BOS  NYK   W  30.52  0.778   4    6  0.667  0.000    2
4
5  Rui Hachimura  LAL  MIN   W  35.08  0.500   1    4  0.250  0.750    3
1
7    Rudy Gobert  MIN  LAL   L  35.33  0.625   0    0  0.000  0.750    3
2

   stl  blk  tov  pf        data  target_win
0    1    1    1   1  2024-10-22           1
1    1    3    1   1  2024-10-22           1
3    1    0    0   2  2024-10-22           1
5    2    1    0   2  2024-10-22           1
7    0    1    1   4  2024-10-22           0
```

# Project 3

In [15]:
```
if 'df_cleaned' in globals():
    p3_df = df_cleaned.copy()
elif 'df' in globals():
    p3_df = df.copy()
else:
    raise NameError("No dataframe found from Project 2 (expected 'df_cleanec

print("Project 3 starting dataframe shape:", p3_df.shape)


# 1) Processing Strings

# I will process string columns only if they exist in this dataset.
string_cols = p3_df.select_dtypes(include=['object', 'string']).columns.toli

if len(string_cols) == 0:
    print("Processing Strings not applied: this dataset does not currently h
else:
    for col in string_cols:
        p3_df[col] = p3_df[col].astype(str).str.strip()


    first_str_col = string_cols[0]
    p3_df[first_str_col + "_lower"] = p3_df[first_str_col].str.lower()
    p3_df[first_str_col + "_char_count"] = p3_df[first_str_col].str.len()

    print("Processing Strings completed on column(s):", string_cols)
    print(p3_df[[first_str_col, first_str_col + "_lower", first_str_col + "_
```

```
Project 3 starting dataframe shape: (4892, 18)
Processing Strings completed on column(s): ['player', 'tm', 'opp', 'res', 'd
ata']
          player   player_lower  player_char_count
0   Jayson Tatum   jayson tatum                 12
1  Anthony Davis  anthony davis                 13
3   Jrue Holiday   jrue holiday                 12
5  Rui Hachimura  rui hachimura                 13
7    Rudy Gobert    rudy gobert                 11
```

In [16]:
```
# 2) Combining and Merging Datasets

# This project uses one main file, so a real merge/combine step is not neces
print("Combining/Merging not required: this project uses one main dataset fi
```

```
Combining/Merging not required: this project uses one main dataset file, so
no second dataset is available to merge.
```

In [17]:
```
# 3) Data Wrangling: Reshaping and Pivoting

numeric_cols = p3_df.select_dtypes(include=[np.number]).columns.tolist()

if len(numeric_cols) < 2:
```

```
    print("Reshaping/Pivoting not fully applied: need at least 2 numeric col
else:
    preferred_id_cols = ['Player', 'player', 'Name', 'name', 'Team', 'team',
    id_col = None
    for c in preferred_id_cols:
        if c in p3_df.columns:
            id_col = c
            break

    if id_col is None:
        p3_df = p3_df.reset_index().rename(columns={'index': 'row_id'})
        id_col = 'row_id'

    value_cols = numeric_cols[:3]

    long_df = p3_df[[id_col] + value_cols].melt(
        id_vars=id_col,
        value_vars=value_cols,
        var_name='metric',
        value_name='value'
    )
    print("Melted (long) dataframe sample:")
    print(long_df.head())

    pivot_df = long_df.pivot_table(
        index='metric',
        values='value',
        aggfunc=['mean', 'median', 'count']
    )
    print("Pivot table summary:")
    print(pivot_df)
```

```
Melted (long) dataframe sample:
          player metric  value
0   Jayson Tatum     mp  30.30
1  Anthony Davis     mp  37.58
2   Jrue Holiday     mp  30.52
3  Rui Hachimura     mp  35.08
4    Rudy Gobert     mp  35.33
Pivot table summary:
             mean  median count
            value   value value
metric
3p       2.165372   2.000  4892
fgpct    0.482701   0.474  4892
mp      34.864025  34.475  4892
```

# Project 4

In [18]:
```
def _pick_any_df():
    dfs = [(k, v) for k, v in globals().items() if isinstance(v, pd.DataFram
    if not dfs:
        raise NameError("No pandas DataFrame found in memory. Run the data l
```

```
    name, df_ = max(dfs, key=lambda kv: kv[1].shape[0] * kv[1].shape[1])
    return df_.copy(), name

viz_df, viz_df_name = _pick_any_df()
print(f"Using dataframe for plots: {viz_df_name} with shape {viz_df.shape}")

TARGET_COL = None

def _infer_target(df):

    hints = ["win", "wins", "result", "outcome", "target", "label", "won", "
    cols = list(df.columns)

    ranked = sorted(cols, key=lambda c: (0 if any(h in str(c).lower() for h
    for c in ranked:
        s = df[c].dropna()
        if s.empty:
            continue
        nunique = s.nunique()
        if nunique == 2:
            return c
    return None

if TARGET_COL is None:
    TARGET_COL = _infer_target(viz_df)

if TARGET_COL is None:
    raise NameError("Could not auto-detect a binary target column. Set TARGE

print("Target column:", TARGET_COL)
print(viz_df[TARGET_COL].value_counts(dropna=False))

target_series = viz_df[TARGET_COL]
if target_series.dtype.name == "category":
    target_series = target_series.astype(str)

vals = [v for v in target_series.dropna().unique()]
if len(vals) != 2:
    raise ValueError(f"Target column {TARGET_COL} does not have exactly 2 ur

map01 = {vals[0]: 0, vals[1]: 1}
y01 = target_series.map(map01)

if y01.isna().mean() > 0.25:
    y01 = pd.Series(pd.factorize(target_series)[0], index=viz_df.index)
    y01 = (y01 == y01.max()).astype(int)
```

```
Using dataframe for plots: df_significant_minutes with shape (13790, 25)
Target column: Res
Res
L    6934
W    6856
Name: count, dtype: int64
```

In [19]:

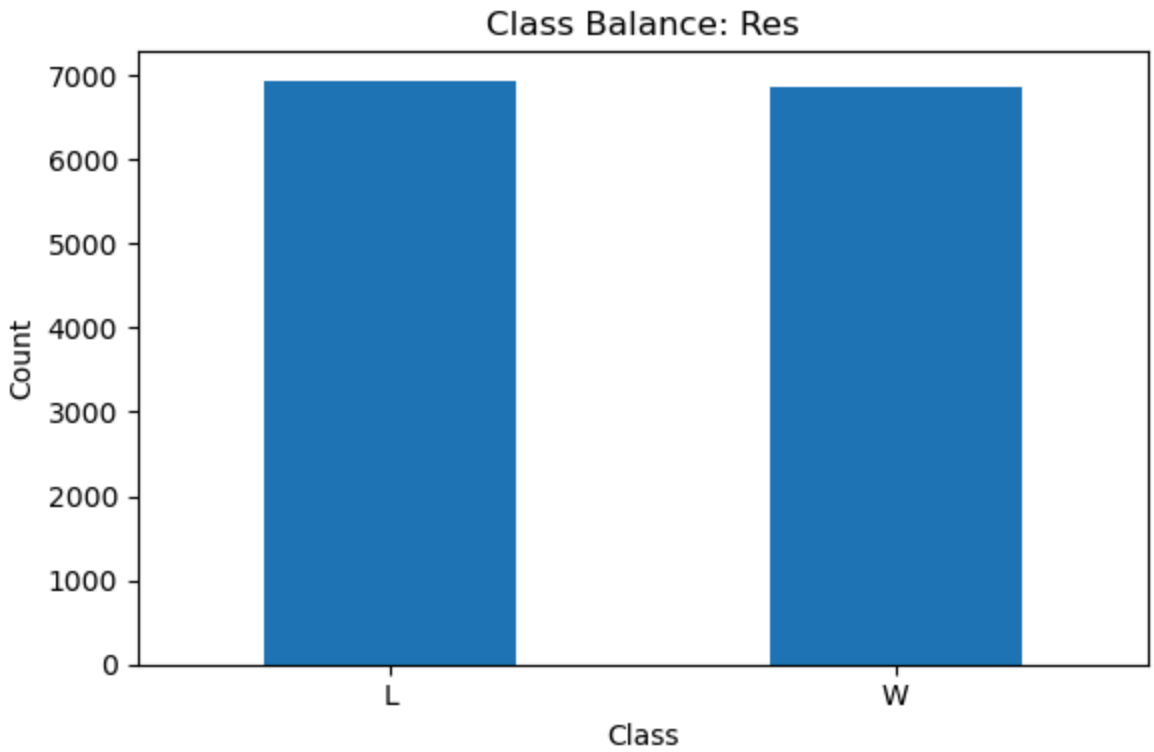
```
num_cols = viz_df.select_dtypes(include=np.number).columns.tolist()
if len(num_cols) == 0:
```

```
    raise ValueError("No numeric columns found to plot.")

tmp = viz_df[num_cols].copy()
tmp["_target01"] = y01
corr = tmp.corr(numeric_only=True)["_target01"].drop("_target01").abs().sort

id_like = {"id", "index", "playerid", "matchid", "gameid"}
top_features = [c for c in corr.index if str(c).lower() not in id_like][:10]
if len(top_features) == 0:
    top_features = corr.index.tolist()[:10]

best_feature = top_features[0]
print("Top features by |corr with outcome|:")
print(corr.loc[top_features].head(10))

corr_pred = viz_df[num_cols].corr(numeric_only=True).abs()
upper = corr_pred.where(np.triu(np.ones(corr_pred.shape), k=1).astype(bool))
pair = upper.stack().sort_values(ascending=False)
best_pair = None
if not pair.empty:
    best_pair = pair.index[0]  # (colA, colB)
    best_pair_r = pair.iloc[0]
    print("Most correlated numeric pair:", best_pair, " |r|=", round(float(b
else:
    print("Not enough numeric columns to compute predictor-pair correlation.
```

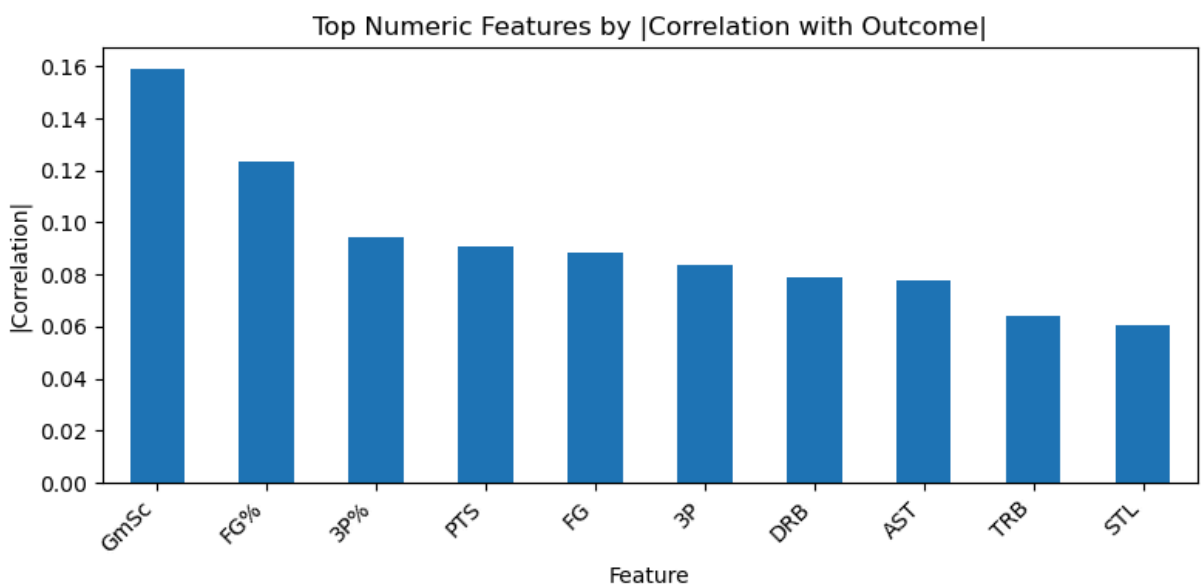
```
Top features by |corr with outcome|:
GmSc    0.159147
FG%     0.123686
3P%     0.094625
PTS     0.091020
FG      0.088346
3P      0.083651
DRB     0.078682
AST     0.077763
TRB     0.064369
STL     0.060529
Name: _target01, dtype: float64
Most correlated numeric pair: ('FG', 'PTS')  |r|= 0.961
```

In [20]:

```
plt.figure(figsize=(6,4))
viz_df[TARGET_COL].value_counts(dropna=False).plot(kind="bar")
plt.title(f"Class Balance: {TARGET_COL}")
plt.xlabel("Class")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

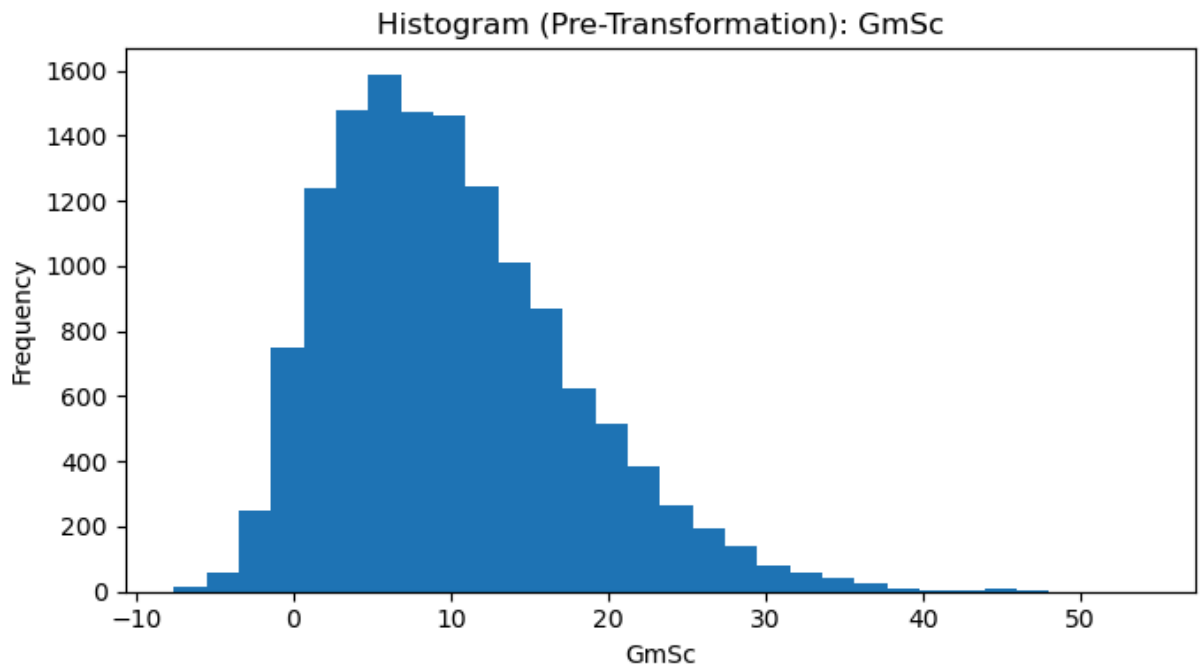



In [21]:
```
plt.figure(figsize=(8,4))
corr.loc[top_features[:10]].sort_values(ascending=False).plot(kind="bar")
plt.title("Top Numeric Features by |Correlation with Outcome|")
plt.xlabel("Feature")
plt.ylabel("|Correlation|")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()

print("Interpretation: Higher bars suggest features more likely to help prec
```

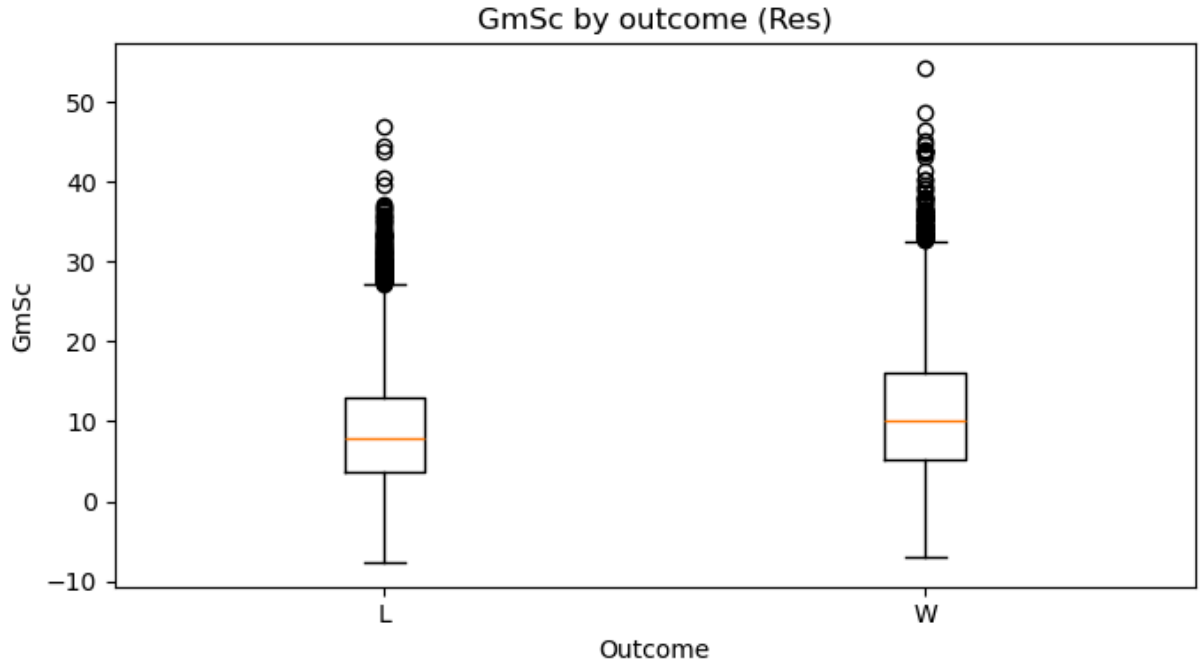



```
Interpretation: Higher bars suggest features more likely to help predict the
outcome. Very low bars may indicate weak predictors.
```

In [22]:
```
plt.figure(figsize=(7,4))
plt.hist(viz_df[best_feature].dropna(), bins=30)
plt.title(f"Histogram (Pre-Transformation): {best_feature}")
plt.xlabel(best_feature)
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

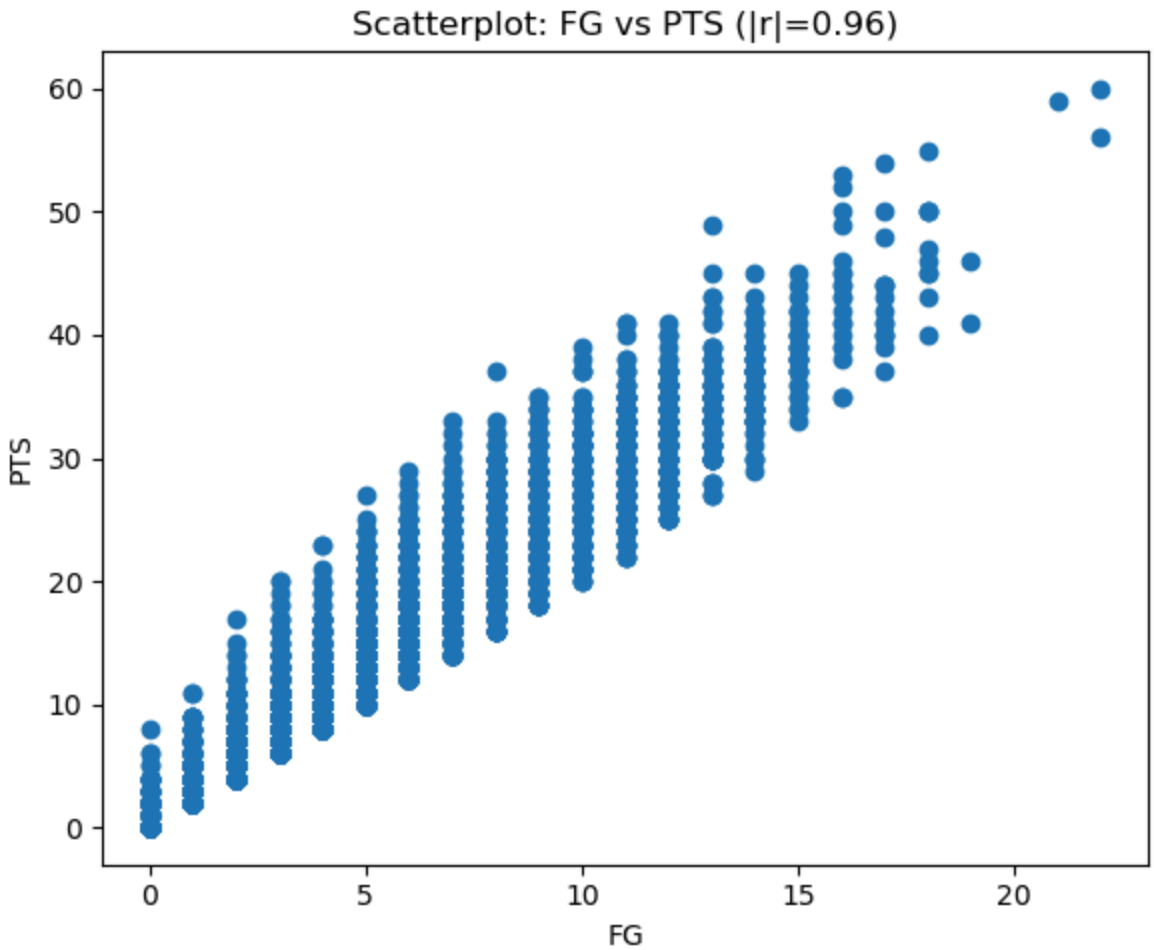



In [23]:
```
labels = list(viz_df[TARGET_COL].dropna().unique())
labels = sorted(labels, key=lambda x: str(x))

data = [viz_df.loc[viz_df[TARGET_COL] == lab, best_feature].dropna() for lab

plt.figure(figsize=(7, 4))
plt.boxplot(data, tick_labels=[str(l) for l in labels])
plt.title(f"{best_feature} by outcome ({TARGET_COL})")
plt.xlabel("Outcome")
plt.ylabel(best_feature)
plt.tight_layout()
plt.show()

print("Interpretation: If medians/IQR differ clearly across classes, the fea
```




```
Interpretation: If medians/IQR differ clearly across classes, the feature is
useful. If they overlap heavily, it may add little value.
```

In [24]:
```
if best_pair is not None:
    a, b = best_pair
    plt.figure(figsize=(6,5))
    plt.scatter(viz_df[a], viz_df[b])
    plt.title(f"Scatterplot: {a} vs {b} (|r|={best_pair_r:.2f})")
    plt.xlabel(a)
    plt.ylabel(b)
    plt.tight_layout()
    plt.show()

    print("Interpretation: A tight linear pattern indicates redundancy.")
else:
    print("No predictor pair scatterplot created because there were not enou
```




```
Interpretation: A tight linear pattern indicates redundancy.
```

# Project 5

In [25]:
```
df9 = viz_df.copy()

num_cols9 = df9.select_dtypes(include=np.number).columns.tolist()
cat_cols9 = df9.select_dtypes(exclude=np.number).columns.tolist()

print("Numeric columns:", num_cols9[:10], "..." if len(num_cols9)>10 else ""
print("Categorical/object columns:", cat_cols9[:10], "..." if len(cat_cols9)

# 1) Grouping Data (Split-Apply-Combine)
group_cols = [TARGET_COL]
agg_features = [best_feature] + [c for c in top_features[1:3] if c in num_cc
agg_features = [c for c in agg_features if c in num_cols9]

if len(agg_features) == 0:
    print("No numeric features available for group statistics.")
else:
    grouped_stats = df9.groupby(TARGET_COL)[agg_features].agg(["mean", "std"
    print("\nGrouped summary by outcome (mean/std/count):")
    display(grouped_stats)

# 2) Applying Functions to Grouped Data
```

```
if best_feature in df9.columns:
    top_by_class = (
        df9.sort_values(best_feature, ascending=False)
           .groupby(TARGET_COL, as_index=False)
           .head(3)
    )
    print(f"\nTop 3 rows within each {TARGET_COL} group by {best_feature}:")
    display(top_by_class[[TARGET_COL, best_feature] + ([c for c in cat_cols9
else:
    print("Skipping top-N within groups because best_feature not found.")

# 3) Pivot Tables (only if categorical exists)
other_cats = [c for c in cat_cols9 if c != TARGET_COL]
if len(other_cats) == 0:
    print("\nPivot Table: Not applicable — no categorical columns besides th
else:
    pivot_cat = other_cats[0]
    if len(agg_features) == 0:
        print("\nPivot Table: Not applicable — no numeric measure found for
    else:
        pivot_val = agg_features[0]
        pivot = df9.pivot_table(index=pivot_cat, columns=TARGET_COL, values=
        print(f"\nPivot Table: mean({pivot_val}) by {pivot_cat} and {TARGET_
        display(pivot)

# 4) Cross-Tabulation (frequency table)
if len(other_cats) == 0:
    print("\nCrosstab: Not applicable — no categorical columns besides the c
else:
    ct_cat = other_cats[0]
    ctab = pd.crosstab(df9[ct_cat], df9[TARGET_COL], margins=True)
    print(f"\nCrosstab: {ct_cat} x {TARGET_COL} (counts):")
    display(ctab)


    pos_label = vals[1] if len(vals)==2 else labels[-1]
    if pos_label in ctab.columns and "All" in ctab.columns:
        rate = (ctab[pos_label] / ctab["All"]).drop("All", errors="ignore")
        rate_df = rate.reset_index()
        rate_df.columns = [ct_cat, "positive_rate"]
        print(f"\nWithin-category positive rate (P({TARGET_COL}={pos_label}
        display(rate_df)
```

```
Numeric columns: ['MP', 'FG', 'FGA', 'FG%', '3P', '3PA', '3P%', 'FT', 'FTA',
'FT%'] ...
Categorical/object columns: ['Player', 'Tm', 'Opp', 'Res', 'Data']

Grouped summary by outcome (mean/std/count):
```

| | GmSc | | | FG% | | | 3P% | | |
|---|---|---|---|---|---|---|---|---|---|
| | mean | std | count | mean | std | count | mean | std | coun |
| **Res** | | | | | | | | | |
| **L** | 8.848414 | 7.088593 | 6934 | 0.434324 | 0.210372 | 6934 | 0.268571 | 0.267695 | 693 |
| **W** | 11.290709 | 8.038383 | 6856 | 0.486582 | 0.208895 | 6856 | 0.321143 | 0.285237 | 685 |

```
Top 3 rows within each Res group by GmSc:
```

| | Res | GmSc | Player | Tm |
|---|---|---|---|---|
| **3619** | W | 54.2 | Giannis Antetokounmpo | MIL |
| **8125** | W | 48.7 | Luka Dončić | DAL |
| **3864** | L | 47.0 | De'Aaron Fox | SAC |
| **3337** | W | 46.5 | Shai Gilgeous-Alexander | OKC |
| **8279** | L | 44.5 | Jimmy Butler | MIA |
| **7368** | L | 43.8 | Nikola Jokić | DEN |

```
Pivot Table: mean(GmSc) by Player and Res:
```

| Res | L | W |
|---|---|---|
| **Player** | | |
| **A.J. Green** | 5.582353 | 5.816667 |
| **AJ Johnson** | 9.200000 | NaN |
| **Aaron Gordon** | 10.360000 | 11.368421 |
| **Aaron Holiday** | 2.288889 | 5.380000 |
| **Aaron Nesmith** | 4.466667 | 7.787500 |
| **...** | ... | ... |
| **Zach Edey** | 9.253846 | 10.086957 |
| **Zach LaVine** | 16.203846 | 18.700000 |
| **Zeke Nnaji** | NaN | 4.925000 |
| **Ziaire Williams** | 5.730435 | 9.306667 |
| **Zion Williamson** | 17.009091 | 22.225000 |

510 rows × 2 columns

```
Crosstab: Player x Res (counts):
```

| Res<br>Player | L | W | All |
|---|---|---|---|
| **A.J. Green** | 17 | 24 | 41 |
| **AJ Johnson** | 1 | 0 | 1 |
| **Aaron Gordon** | 10 | 19 | 29 |
| **Aaron Holiday** | 9 | 10 | 19 |
| **Aaron Nesmith** | 6 | 8 | 14 |
| **...** | ... | ... | ... |
| **Zach LaVine** | 26 | 18 | 44 |
| **Zeke Nnaji** | 0 | 4 | 4 |
| **Ziaire Williams** | 23 | 15 | 38 |
| **Zion Williamson** | 11 | 4 | 15 |
| **All** | 6934 | 6856 | 13790 |

511 rows × 3 columns

```
Within-category positive rate (P(Res=L | Player)):
```

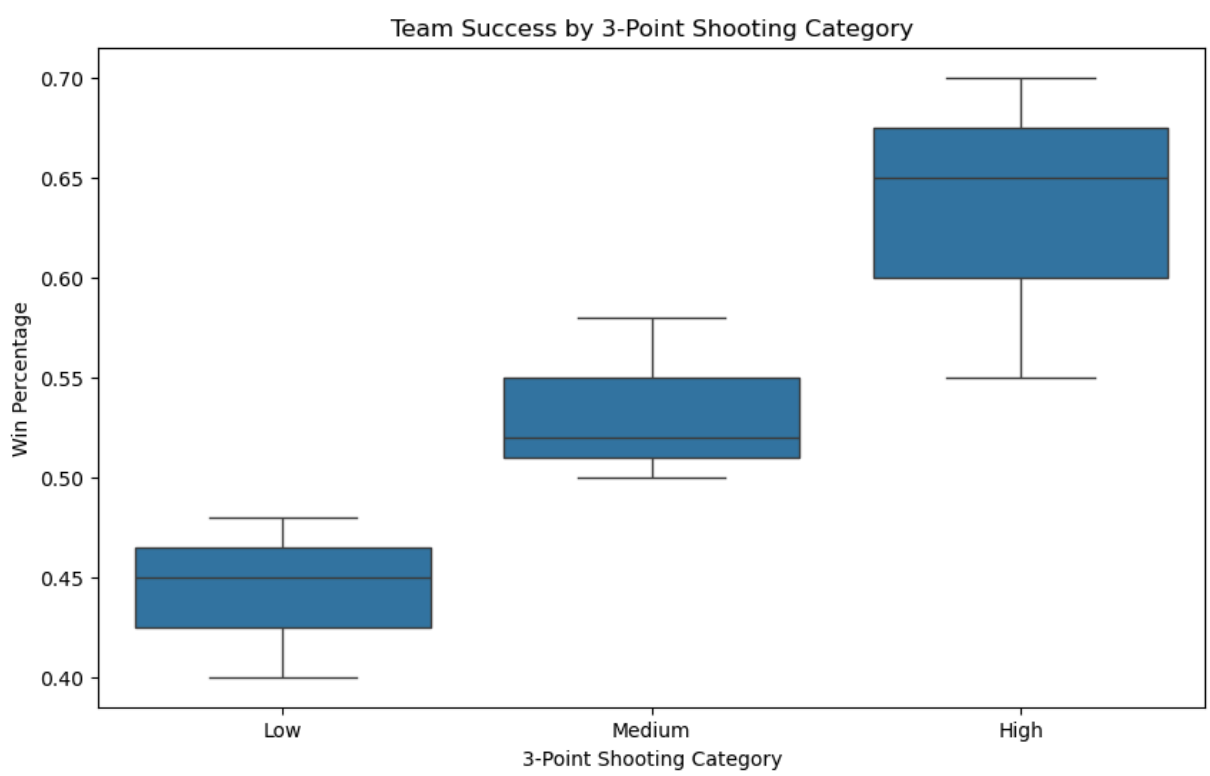
| | Player | positive_rate |
|---|---|---|
| **0** | A.J. Green | 0.414634 |
| **1** | AJ Johnson | 1.000000 |
| **2** | Aaron Gordon | 0.344828 |
| **3** | Aaron Holiday | 0.473684 |
| **4** | Aaron Nesmith | 0.428571 |
| **...** | ... | ... |
| **505** | Zach Edey | 0.361111 |
| **506** | Zach LaVine | 0.590909 |
| **507** | Zeke Nnaji | 0.000000 |
| **508** | Ziaire Williams | 0.605263 |
| **509** | Zion Williamson | 0.733333 |

510 rows × 2 columns

# Project 6

In [26]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

nba_data = pd.DataFrame({
    'team': ['Team A', 'Team B', 'Team C', 'Team D', 'Team E', 'Team F', 'Te
    'three_pt_percentage': [0.32, 0.36, 0.40, 0.33, 0.38, 0.35, 0.42, 0.31,
    'win_percentage': [0.45, 0.52, 0.65, 0.48, 0.55, 0.50, 0.70, 0.40, 0.58]
})

nba_data['three_pt_category'] = pd.qcut(nba_data['three_pt_percentage'], 3,
                                        labels=['Low', 'Medium', 'High'])

low_group = nba_data[nba_data['three_pt_category'] == 'Low']['win_percentage
medium_group = nba_data[nba_data['three_pt_category'] == 'Medium']['win_perc
high_group = nba_data[nba_data['three_pt_category'] == 'High']['win_percenta

f_stat, p_value = stats.f_oneway(low_group, medium_group, high_group)
print(f"ANOVA results: F-statistic = {f_stat}, p-value = {p_value}")

plt.figure(figsize=(10, 6))
sns.boxplot(x='three_pt_category', y='win_percentage', data=nba_data)
plt.title('Team Success by 3-Point Shooting Category')
plt.xlabel('3-Point Shooting Category')
plt.ylabel('Win Percentage')
plt.show()
```

```
ANOVA results: F-statistic = 8.836956521739141, p-value = 0.0162795953942857
3
```




In [31]:
```
model.summary()
```

Out[31]:

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | target_win | **R-squared:** | 0.040 |
| **Model:** | OLS | **Adj. R-squared:** | 0.038 |
| **Method:** | Least Squares | **F-statistic:** | 22.35 |
| **Date:** | Fri, 20 Mar 2026 | **Prob (F-statistic):** | 1.23e-37 |
| **Time:** | 00:02:42 | **Log-Likelihood:** | -3451.5 |
| **No. Observations:** | 4892 | **AIC:** | 6923. |
| **Df Residuals:** | 4882 | **BIC:** | 6988. |
| **Df Model:** | 9 | | |
| **Covariance Type:** | nonrobust | | |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **const** | 0.5331 | 0.075 | 7.070 | 0.000 | 0.385 | 0.681 |
| **3p** | 0.0561 | 0.006 | 8.933 | 0.000 | 0.044 | 0.068 |
| **3pa** | -0.0260 | 0.003 | -7.679 | 0.000 | -0.033 | -0.019 |
| **orb** | -0.0002 | 0.005 | -0.040 | 0.968 | -0.009 | 0.009 |
| **ast** | 0.0195 | 0.002 | 7.933 | 0.000 | 0.015 | 0.024 |
| **stl** | 0.0223 | 0.006 | 3.649 | 0.000 | 0.010 | 0.034 |
| **blk** | 0.0361 | 0.007 | 5.089 | 0.000 | 0.022 | 0.050 |
| **tov** | -0.0221 | 0.004 | -5.003 | 0.000 | -0.031 | -0.013 |
| **pf** | -0.0203 | 0.005 | -4.013 | 0.000 | -0.030 | -0.010 |
| **mp** | -0.0012 | 0.002 | -0.521 | 0.602 | -0.006 | 0.003 |

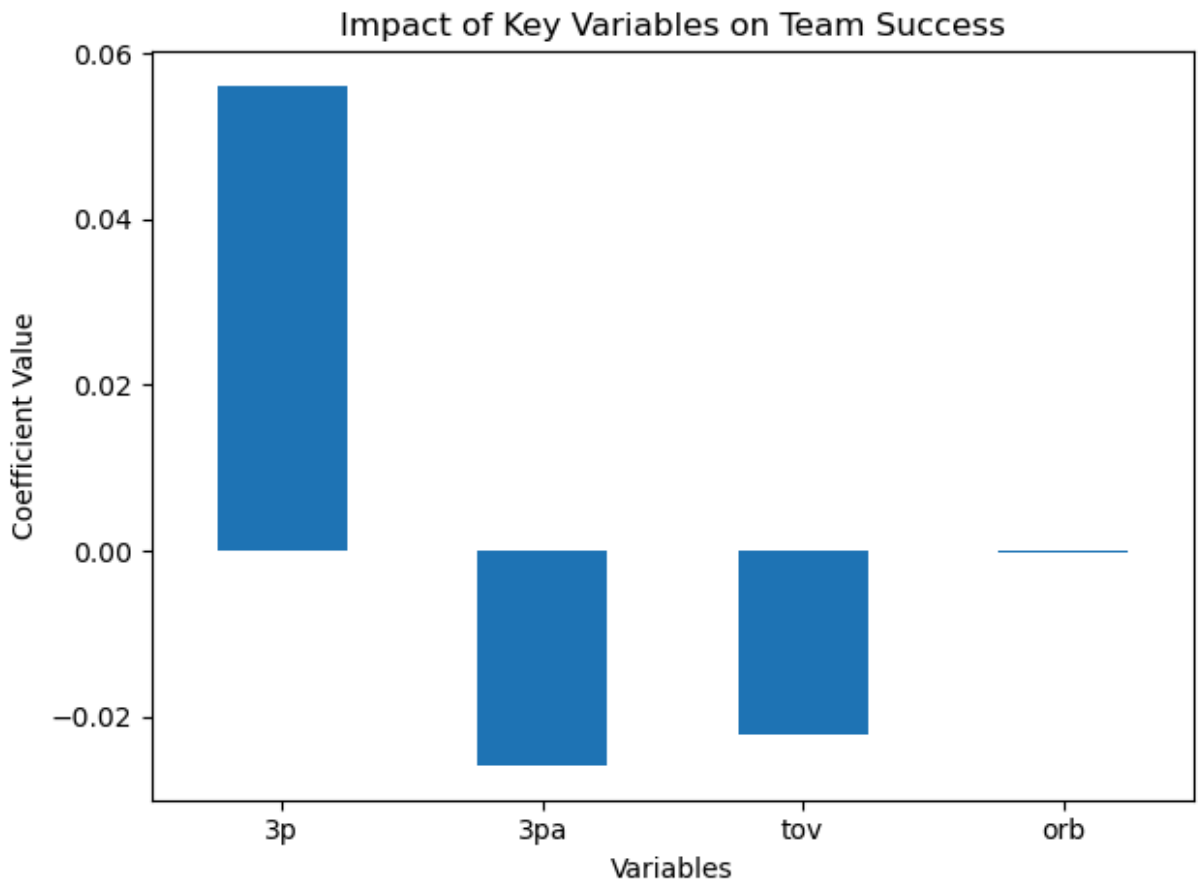
| | | | |
|---|---|---|---|
| **Omnibus:** | 18833.681 | **Durbin-Watson:** | 2.063 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 693.871 |
| **Skew:** | -0.019 | **Prob(JB):** | 2.13e-151 |
| **Kurtosis:** | 1.155 | **Cond. No.** | 388. |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

In [33]:

```
coefficients = model.params[['3p', '3pa', 'tov', 'orb']]

coefficients.plot(kind='bar')

plt.title("Impact of Key Variables on Team Success")
```

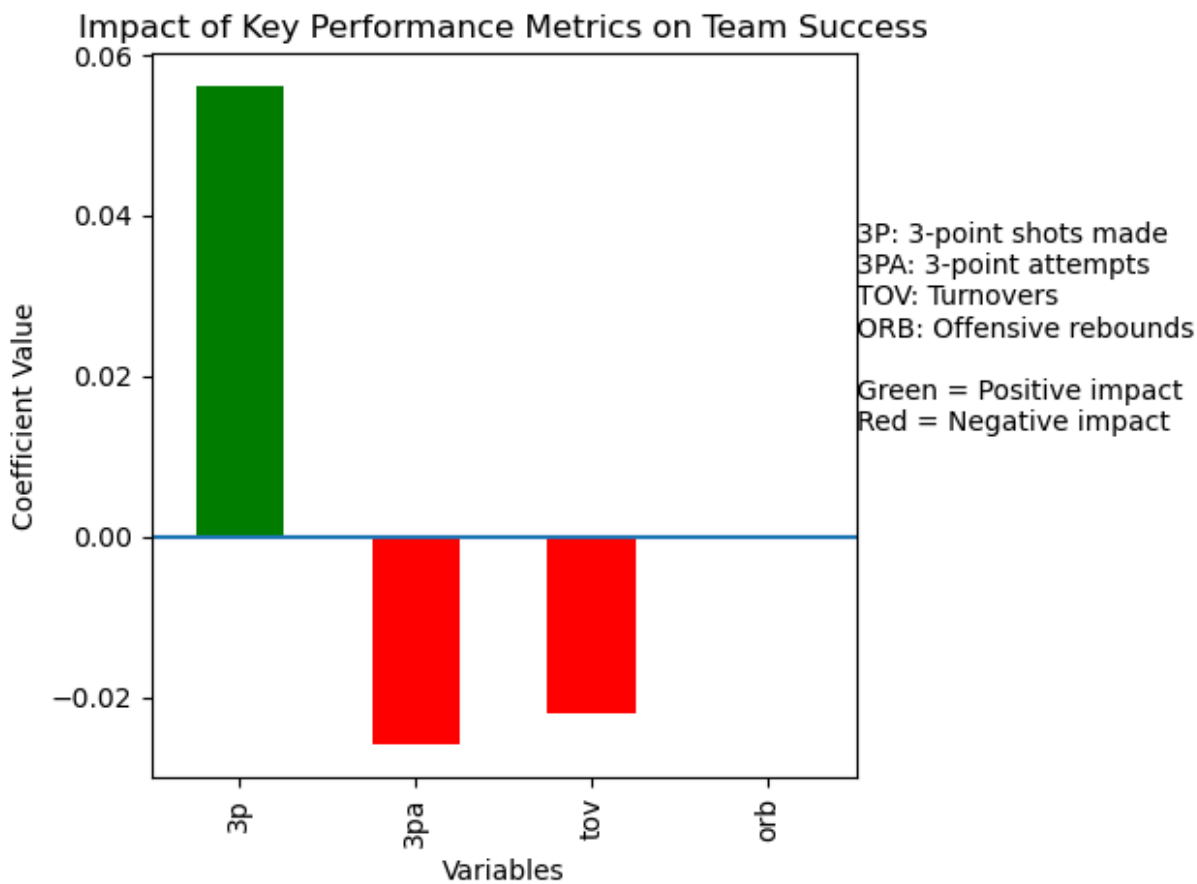
```
plt.ylabel("Coefficient Value")
plt.xlabel("Variables")
plt.xticks(rotation=0)
plt.tight_layout()

plt.show()
```




In [35]:
```
results = pd.DataFrame({
    "Coefficient": model.params[['3p','3pa','tov','orb']],
    "P-value": model.pvalues[['3p','3pa','tov','orb']]
})
results
```

Out[35]:

| | Coefficient | P-value |
|---|---|---|
| **3p** | 0.056066 | 5.750138e-19 |
| **3pa** | -0.025951 | 1.921189e-14 |
| **tov** | -0.022112 | 5.831720e-07 |
| **orb** | -0.000190 | 9.679017e-01 |

In [47]:
```
vars = ['3p', '3pa', 'tov', 'orb']

coefs = model.params[vars]

coefs.plot(kind='bar')
```

```
plt.title("Impact of Key Performance Metrics on Team Success")
plt.ylabel("Coefficient Value")
plt.xlabel("Variables")
plt.xticks(rotation=0)

colors = ['green' if x > 0 else 'red' for x in coefs]

coefs.plot(kind='bar', color=colors)

plt.axhline(0)

text = (
    "3P: 3-point shots made\n"
    "3PA: 3-point attempts\n"
    "TOV: Turnovers\n"
    "ORB: Offensive rebounds\n\n"
    "Green = Positive impact\n"
    "Red = Negative impact"
)

plt.text(3.5, max(coefs)*0.7, text, fontsize=10, verticalalignment='top')

plt.tight_layout()
plt.show()
```